

Artem Semidetnov

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Education

BSc	Saint-Petersburg State University , Mathematics	2021 – 2025
BSc	Neapolis University Of Pafos , Applied Computer Science JetBrains Program	2022 – 2024
MSc	Geneva University , Mathematics	2025 – 2027

Research Interests

- Categorical homotopy theory; simplicial methods and operads; homotopy type theory and computer-formalized approaches; geometric and combinatorial group theory (stable commutator length, one-relator groups).

Publications

The operad associated to a crossed simplicial group ↗	Dec 2025
Artem Semidetnov	
https://doi.org/10.48550/arXiv.2512.14498 ↗	
On the geometry of free nilpotent groups ↗	May 2021
Artem Semidetnov , Ruslan Magdiev	
https://doi.org/10.48550/arXiv.2106.00095 ↗	

AI-Assisted Mathematical Research

On Some More Problems from the Kourovka Notebook ↗	Aug 2026
Vasily Ionin, Artem Semidetnov	
https://doi.org/10.48550/arXiv.2608.29219 ↗	
The stable commutator length of a relator is not a one-relator group invariant ↗	Aug 2026
Artem Semidetnov	
https://doi.org/10.48550/arXiv.2608.21465 ↗	

Work Experience

JetBrains , Research Intern in HoTT and Dependent Types Lab	Remote
• Developed AI features for the Arend language including MCP servers for typechecking, IDE integration. Contributed to the Arend plugin in IntelliJ IDEA. • Developed an experimental LLM coding agent for writing Arend code using framework Koog. • Fine-tuned models for Arend code completion.	Nov 2025 – Apr 2026
JetBrains , Research Intern in HoTT and Dependent Types Lab	Pafos, Cyprus
• I was developing the official library in the Arend language. I formalized different results in algebra and homotopy type theory, including following. • Formalized Eckmann-Hilton argument, Eilenberg-MacLane spaces, Homologies of types. • Formalized automorphisms of groups, Schur's Lemma, Maschke's Lemma, Group actions characterizations. (Some of these results are in the 1.10 release ↗).	June 2024 – Sept 2024
IPONWEB (acquired by Criteo) , Machine Learning Intern	Pafos, Cyprus
• Criteo has a ML tool that analyzes sites and produces word2vec representations. Reverse-engineered the behaviour of this tool and analyzed its possible applications.	May 2023 – Sept 2023

Awards and Scholarships

Scholarship by "Rodnye Goroda" (a social investment program of PJSC "Gazprom neft"), 2023-2024

JetBrains Scholarship in Neapolis University

Travel grant of Chebyshev laboratory for [Winter student school in mathematics and CS of HSE and MCS](#), Moscow

[Excellence Master Fellowship at Geneva University](#)

Teaching Experience

Sirius educational center, Teaching assistant on the course "[Groups of intermediate growth](#)" [↗](#) Sochi, Russia
Apr 2024 – May 2024

Laboratory for continuous mathematical education, mathematics teacher for gifted students Saint-Petersburg, Russia
2021 – 2023

Mathematics and Computer Science faculty program for prospective students, Teaching assistant on the course Braid and knot theory Saint-Petersburg, Russia
June 2023

Selected Talks

On the geometry of free nilpotent groups Novosibirsk, Russia
• Siberian summer conference 2021

"On the Poisson boundary of lamplighter groups" [↗](#) Saint-Petersburg, Russia
• St. Petersburg Seminar on Representation Theory and Dynamical Systems

"Twisting numbers on braid and Thompson's groups" [↗](#) N.Novgorod, Russia
• Topological Methods in Dynamics and Related Topics VII 2024

"Plus construction and simplicial methods" [↗](#) Pushkin, Russia
• Conference "Algebraic Topology, geometry, combinatorics, and data analysis" [↗](#) Aug 2025

"The operad associated to a crossed simplicial group" [↗](#) Copenhagen, Denmark
• Young Topologists Meeting 2026 July 2026

Schools and Workshops

Teaching Higher Category Theory with Computers [↗](#) ICERM, Brown University, Providence, RI
• Topical workshop on directed HoTT and computer-formalized higher category theory, organized by Emily Riehl, Nima Rasekh, Jonathan Weinberger. Participant (travel and lodging funded). Aug 2026

Miscellaneous

- Finalist of 2020 Intel ISEF
- Intel ISEF alumni
- Winner of 2019 Baltic SEF, PDMI special prize in 2019 Baltic SEF
- 3rd team place in 2019 [International Tournament of Young Mathematicians](#) [↗](#) in Barcelona, Spain
- Organizer of the [Euler International Mathematical Institute's functional analysis seminar](#) [↗](#)
- English level C1 (IELTS 8.0/9, taken in 2020, 2024)
- Invited judge in Saint-Petersburg Tournament of Young Mathematicians (since 2021)
- Created mathematical problem for 2024 International Tournament of Young Mathematicians [10th in here](#) [↗](#)
- Invited judge in International Tournament of Young Mathematicians 2024, 2025

- A member of Scientific Organization Committee in ETEAM 2025 [↗](#)